

Helmet Protection in E-Scooter Falls

A Finite Element Parametric Study

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Abstract

TBD

NOTE: Everything presented here is very preliminary as this report captures our first set of results. We will continue refining both the text and the simulations as the work progresses. We hope this gives a clear enough picture of our initial progress so far.

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1 Introduction

Electric scooters (e-scooters) have rapidly expanded across global urban centers, providing convenient short-trip transportation while significantly increasing accident and injury rates. In Sweden alone, a recent surge in e-scooter accidents resulted in seven fatalities and 6,624 injuries over a single year, prompting a formal regulatory review [1]. Pai et al. investigated fatal crashes involving e-scooters, e-bikes, and conventional bicycles in Sweden between 2016 and 2024. None of the 18 fatally injured e-scooter riders wore a helmet[2]. Trivedi et al. reported head injuries in 40.2% of 249 patients presenting to two emergency departments with e-scooter-related injuries; 4.4% of injured riders were documented as wearing a helmet [3]. In Stockholm, emergency department data indicates that approximately one accident requiring an emergency department visit per 33,000 e-scooter trips in Stockholm during 2019–2020 [4].

Recent finite element (FE) studies have begun characterizing the severe loading conditions inherent to e-scooter falls. For instance, Wei et al. simulated e-scooter curb collisions and found oblique head-ground impacts in all 27 simulated falls, with 78% of impacts occurring on the forehead [5]. They reported mean vertical and tangential head-ground impact velocities of 5.7 ± 1.5 m/s and 3.7 ± 2.0 m/s, respectively [5]. While standard bicycle helmets significantly reduce linear acceleration and the Head Injury Criterion (HIC) during these falls, they struggle to mitigate angular acceleration, which remains dangerously above severe injury thresholds [5]. Complementary experimental crash tests by Bailly et al. further confirm that standard bicycle helmets offer insufficient protection against the complex rotational kinematics unique to e-scooter crashes [6].

These findings suggest that standard bicycle helmets—which are certified using purely linear drops at lower impact speeds—may not provide adequate protection for e-scooter riders. To address this gap, this project aims to evaluate how specific helmet design parameters influence head injury metrics under literature-informed e-scooter fall conditions. Utilizing an FE parametric study with a Hybrid III dummy, simplified brain, and simplified helmet model, we prioritize computational efficiency to systematically isolate key variables. Based on this high prevalence of forehead impacts [5], this initial study focuses exclusively on frontal loading configurations.

Our parametric sweep focuses on two primary variables:

- **Impact Velocity:** The vertical and tangential

impact velocity components were each varied independently across three literature-derived values (mean minus one standard deviation, mean, and mean plus one standard deviation) reported by Wei et al. [5], yielding nine velocity combinations in total.

- **Helmet Liner Stiffness:** Systematically varied by scaling the stress-strain compression curve (SFO parameter) of the foam liner to assess energy absorption capabilities.

To comprehensively assess brain injury risk, this study extracts both global kinematic metrics (e.g., HIC₃₆ for focal injury risk, BrIC for rotational injury risk) and local tissue-level responses (e.g., maximum 1st principal Green strain in the brain hemispheres and falx). By characterizing which types of head injuries occur under specific e-scooter loading conditions, this research aims to identify which protective design parameters most effectively reduce the severity of traumatic brain injuries.

2 Methods

2.1 Baseline Model

Simulations were performed using the finite element model of a Hybrid III 50th-percentile dummy provided for the HL2035 course, comprising a rigid upper-body skeleton, a scalp, and a simplified brain represented by two hemispheres separated by a falx. A simplified helmet model, consisting of an outer shell and an inner foam liner, and a rigid flat ground surface were included to represent the head-ground impact phase of an e-scooter fall. All simulations were run in LS-DYNA (SI units: meters, seconds, kilograms), using the course-provided `Main_File.k` as the baseline input deck, with `Helmet.k` and `Ground.k` included or excluded depending on the case being simulated.

Rather than modeling the full pre-impact fall trajectory (curb collision, body rotation, and initial ground contact with the limbs, as simulated by Wei et al. [5]), this study applies literature-derived head-ground impact velocities directly as the initial condition for the dummy, consistent with our project’s focus on the head-ground impact phase itself.

2.2 Parameter 1: Impact Velocity

Following Wei et al.[5], the head-ground impact velocity was decomposed into a vertical component (perpendicular to the ground) and a tangential component (parallel to the ground), reflecting oblique impacts in their simulated falls. The vertical and tan-

gential components were each varied independently across three literature-derived values, corresponding to the mean minus one standard deviation, the mean, and the mean plus one standard deviation of the values reported by Wei et al.[5] (vertical: 5.7 ± 1.5 m/s; tangential: 3.7 ± 2.0 m/s), yielding nine velocity combinations:

- Vertical: 4.2, 5.7, 7.2 m/s
- Tangential: 1.7, 3.7, 5.7 m/s

The sweep uses the pooled statistics reported by Wei et al. across their simulated falls as a literature-informed basis for a controlled sensitivity analysis. The nine combinations are constructed test conditions, not nine reconstructed accidents or estimates of their occurrence probabilities. Independently varying the two components changes both resultant impact speed and impact angle.

2.3 Parameter 2: Helmet Liner Stiffness

The helmet liner is modeled with *MAT_MODIFIED_HONEYCOMB, using the supplied stress-strain curve to define its compressive response. This response was varied by scaling this curve's stress axis (the SFO parameter), rather than by altering an elastic modulus directly, in line with established practice for this material model. The curve's strain axis, plateau shape, shear response, and density were held at their baseline values, isolating compressive stress magnitude as a single controlled variable.

The baseline liner curve (SFO = 1,000,000, as provided in the course material) was scaled to 0.25 $\times$, 0.5 $\times$, 1 $\times$ (baseline), and 2 $\times$, bracketing the originally provided value with both softer and stiffer liner variants:

- SFO = 250,000 (0.25 $\times$)
- SFO = 500,000 (0.5 $\times$)
- SFO = 1,000,000 (1 $\times$, baseline)
- SFO = 2,000,000 (2 $\times$)

This range was chosen to span a broad range of compressive response levels around the course-provided baseline, allowing the influence of liner response on head injury metrics to be assessed in both directions. The liner's shear curve was not varied in this study; given that e-scooter head-ground impacts are inherently oblique [5], the liner's shear response represents a natural extension of this parameter study and is identified as a candidate for future work.

2.4 Simulation Duration

The course-provided default termination time of 0.2 s was reduced to 0.04 s, and injury metrics were extracted within a 20 ms analysis window measured from the start of the simulation. Two features of the model govern these choices.

First, the head-ground impact is confined to the early part of the simulation. The helmet outer surface is initially separated from the ground by 12.3 mm, giving contact onset at 1.7, 2.2 and 2.9 ms for vertical velocities of 7.2, 5.7 and 4.2 m/s, with the contact event lasting on the order of 15 ms.

Second, the dummy model contains a rigid torso surface that is constrained to the 25 kg thorax rigid body and is included in the ground contact definition. Its leading edge is initially separated from the ground by 148 mm, so under the prescribed initial velocities it reaches the ground at approximately 21, 26 and 35 ms for vertical velocities of 7.2, 5.7 and 4.2 m/s. This secondary torso-ground contact is an artifact of prescribing a uniform initial velocity to the entire dummy rather than simulating the fall trajectory. Because it involves a rigid body with no compliant structure, at the original 0.2 s termination time it produced negative solid-element volumes in the ground and error termination in several high-velocity cases.

The 20 ms analysis window fully contains the head-ground impact while excluding the secondary torso contact in all nine velocity combinations; no results beyond 20 ms were used. The model also contains no gravity load, so the post-impact trajectory carries no physical meaning in any case.

2.5 Data Analysis

For each simulation, global kinematic injury metrics (peak resultant linear acceleration, HIC₃₆, peak angular velocity, peak angular acceleration, and BrIC) will be extracted from the head center-of-mass node output, and local tissue-level metrics (maximum 1st principal Green strain in each brain hemisphere and the falx) will be extracted from the brain elements. These metrics will be compared across the tested velocity combinations and helmet liner stiffness values to assess their relative influence on head injury risk.

3 Results

3.1 Helmet Liner Stiffness and Tangential Velocity Sweep – first part

The results presented in this section correspond to the parameter sweep assigned to one author: vertical

impact velocity held fixed at 4.2 m/s (the lower bound of the three-level sweep described in Section 2), while tangential velocity (1.7, 3.7, and 5.7 m/s) and helmet liner stiffness (SFO = 0.25 $\times$, 0.5 $\times$, 1 $\times$, and 2 $\times$ baseline, plus an unhelmeted control) were varied across all combinations, for a total of 15 simulations. These are partial results from one portion of the group’s full parametric study; the remaining vertical-velocity conditions assigned to other group members, cross-comparisons between all velocity groups, and local (brain strain) injury criteria will be integrated in the final report.

Table 1 summarizes the peak resultant linear acceleration recorded at the head center of mass (Node 8) and its corresponding time of occurrence, for all 15 simulated cases.

Table 1. Peak resultant linear acceleration at Node 8, vertical velocity fixed at 4.2 m/s.

Tang. (m/s)	Helmet Condition	Peak Accel.	Peak Time (s)
1.7	No Helmet	4684.20	0.0081
	SFO 0.25 $\times$	2672.70	0.0064
	SFO 0.5 $\times$	2484.50	0.0063
	SFO 1 $\times$ (base)	2323.53	0.0063
	SFO 2 $\times$	2297.46	0.0057
3.7	No Helmet	5620.21	0.0081
	SFO 0.25 $\times$	2682.67	0.0064
	SFO 0.5 $\times$	2508.00	0.0063
	SFO 1 $\times$ (base)	2371.25	0.0061
	SFO 2 $\times$	2335.59	0.0060
5.7	No Helmet	5102.47	0.0081
	SFO 0.25 $\times$	2664.18	0.0065
	SFO 0.5 $\times$	2436.34	0.0064
	SFO 1 $\times$ (base)	2327.45	0.0062
	SFO 2 $\times$	2287.80	0.0058

Figure 1 compares peak resultant acceleration across all tested helmet conditions, grouped by tangential velocity. All helmeted configurations reduced peak head acceleration by approximately 50% relative to the unhelmeted case, consistently across all three tangential velocities tested. Within the helmeted cases, peak acceleration decreased monotonically as liner stiffness (SFO) increased, from the softest (0.25 $\times$) to the stiffest (2 $\times$) configuration. Tangential velocity itself had comparatively little effect on peak acceleration for a given helmet condition.

Figure 2 shows the full resultant acceleration time history for all 15 cases, grouped into three panels by tangential velocity. The unhelmeted condition consistently produces both a higher initial peak and a more pronounced oscillatory response throughout the simulated window, while all four helmeted configurations converge to a similar response after the initial impact peak.

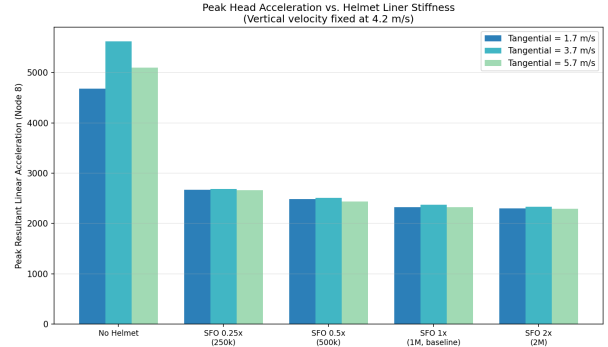


Figure 1. Peak resultant linear acceleration at Node 8 across all tested helmet liner stiffness values, grouped by tangential velocity. Vertical velocity fixed at 4.2 m/s.

3.2 Helmet Liner Stiffness and Tangential Velocity Sweep – second part

This portion of the sweep holds vertical impact velocity fixed at the upper bound of the three-level study, 5.7 m/s, and varies the same tangential velocities (1.7, 3.7, 5.7 m/s) together with four liner compression strengths (0.25 $\times$, 0.5 $\times$, 1 $\times$, 2 $\times$ baseline SFO) plus an unhelmeted control, again yielding 15 cases. Where the previous subsection tracked peak acceleration alone, the results below add the Head Injury Criterion (HIC36) at Node 8, since peak acceleration and HIC do not necessarily move together: HIC integrates the acceleration pulse over a rolling 36 ms window rather than capturing only its instantaneous maximum, and the two metrics can therefore rank the same set of cases differently. Fourteen of the fifteen planned cases are included here; one case is still pending re-verification after a file-handling issue during batch setup and will be finalized for the complete report.

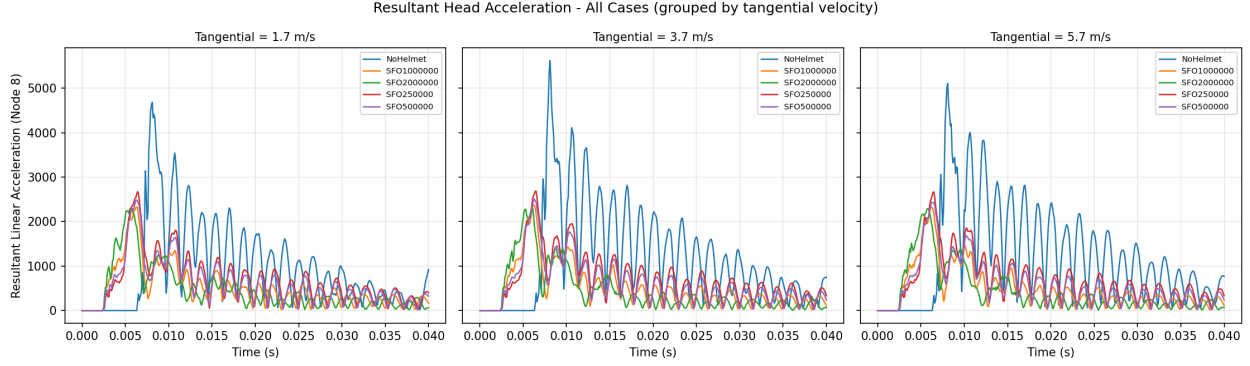


Figure 2. Resultant head acceleration time histories for all 15 simulated cases, grouped by tangential velocity (1.7, 3.7, and 5.7 m/s). Each panel overlays the unhelmeted control and all four tested helmet liner stiffness values (SFO). Vertical velocity fixed at 4.2 m/s.

Table 2. Peak resultant linear acceleration and HIC36 at Node 8, vertical velocity fixed at 5.7 m s⁻¹. All reported cases passed an energy-conservation check (total energy within 0.3% of its initial value throughout).

Tang. (m/s)	Helmet Condition	Peak (g)	HIC36
1.7	No Helmet	850.0	12,706.7
	SFO 0.25×	353.9	3,164.9
	SFO 0.5×	406.1	4,767.8
	SFO 1× (base)	374.5	3,886.1
3.7	No Helmet	742.2	14,960.7
	SFO 0.5×	416.9	5,312.1
	SFO 1× (base)	421.8	5,946.2
	SFO 2×	363.1	2,999.0
5.7	No Helmet	705.4	19,125.5
	SFO 0.25×	389.6	5,529.3
	SFO 0.5×	390.8	4,942.7
	SFO 1× (base)	363.6	3,815.7
	SFO 2×	368.9	3,113.6

across every tangential velocity and liner strength tested. Where the two subsections diverge is in how liner strength interacts with the outcome once the helmet is already on. Rather than a single strength value performing best throughout, fig. 3 shows the optimal SFO shifting with tangential velocity: at 1.7 m/s HIC36 is lowest at the softest liner tested and rises before partially recovering as SFO increases, while at 5.7 m/s the trend reverses, with HIC36 falling steadily as the liner gets stiffer. A plausible explanation is the liner's finite crush travel: a liner too soft for a given impact energy can bottom out (densify) before that energy is absorbed, producing a sharp late-stage force spike, whereas a liner too stiff for the same impact severity transmits more force from the very start of contact. Since impact energy grows with velocity, the strength that balances these two failure modes is not fixed but shifts with how hard the impact actually is.

A modeling limitation surfaced specifically at the highest tangential velocity: several liner variants at 5.7 m/s could not complete the full 200 ms simulation window, terminating early due to negative-volume element failure in the liner mesh, consistent with localized over-crush at this impact severity. These cases were re-run with the termination time shortened to roughly 40 ms – long enough to capture the full acceleration peak and the complete 36 ms HIC window, while avoiding the point of numerical breakdown. This is treated here as a mesh/discretization limitation at high impact severity rather than a modeling error, and is discussed further where relevant.

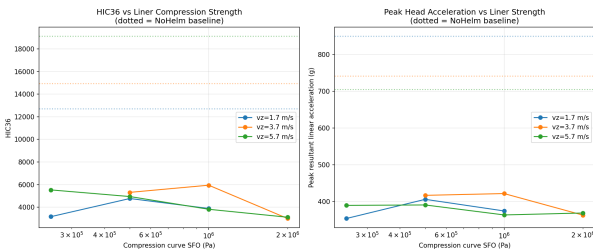


Figure 3. HIC36 and peak resultant linear acceleration as a function of liner compression strength, for each tested tangential velocity, vertical velocity fixed at 5.7 m s⁻¹. Dotted lines mark the no-helmet baseline for each tangential velocity.

As in the previous subsection, the helmet's protective benefit is large and consistent: HIC36 drops by roughly 70–85 % relative to the unhelmeted control

4 Discussion

5 Conclusion

References

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